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In [1]: # -----
# Library Management System
# Python Mini Project
# -----

# List to store library book records (10 books)
library = [
    {"ID": 1, "Title": "Python Programming", "Author": "John Doe", "Category": "PROGRAMMING"},
    {"ID": 2, "Title": "Data Structures", "Author": "Mark Allen", "Category": "COMPUTER SCIENCE"},
    {"ID": 3, "Title": "Machine Learning", "Author": "Andrew Ng", "Category": "AI"},
    {"ID": 4, "Title": "Deep Learning", "Author": "Ian Goodfellow", "Category": "AI"},
    {"ID": 5, "Title": "Operating Systems", "Author": "Silberschatz", "Category": "COMPUTER SCIENCE"},
    {"ID": 6, "Title": "Database Management", "Author": "Elmasri", "Category": "DATABASE"},
    {"ID": 7, "Title": "Computer Networks", "Author": "Tanenbaum", "Category": "NETWORKING"},
    {"ID": 8, "Title": "Web Development", "Author": "Tim Berners-Lee", "Category": "WEB"},
    {"ID": 9, "Title": "Cyber Security", "Author": "Bruce Schneier", "Category": "SECURITY"},
    {"ID": 10, "Title": "Data Science", "Author": "Joel Grus", "Category": "DATA"}
]

# Function to add a new book
def add_book():
    try:
        book_id = int(input("Enter Book ID: "))
        title = input("Enter Book Title: ").strip().title()
        author = input("Enter Author Name: ").strip().title()
        category = input("Enter Book Category: ").strip().upper()

        book = {
            "ID": book_id,
            "Title": title,
            "Author": author,
            "Category": category
        }

        library.append(book)
        print("Book added successfully!")

    except ValueError:
        print("Invalid input! Book ID must be a number.")

```

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# Function to view all books
def view_books():
    if len(library) == 0:
        print("No books available in the library.")
    else:
        print("\n--- Library Book Records ---")
        for book in library:
            print("ID:", book.get("ID"))
            print("Title:", book.get("Title"))
            print("Author:", book.get("Author"))
            print("Category:", book.get("Category"))
            print("-----")

# Function to search books by category
def search_book():
    search_category = input("Enter category to search: ").strip().upper()
    found = False

    for book in library:
        if book.get("Category") == search_category:
            print(book)
            found = True

    if not found:
        print("No books found in this category.")

# Function to save records to file
def save_to_file():
    with open("library_records.txt", "w") as file:
        for book in library:
            file.write(str(book) + "\n")
    print("Library records saved successfully!")

# Menu-driven program
while True:
    print("\n===== Library Management Menu =====")
    print("1. Add Book")
    print("2. View Books")
    print("3. Search Book by Category")
    print("4. Save & Exit")
```

```
choice = input("Enter your choice (1-4): ").strip()

if choice == "1":
    add_book()
elif choice == "2":
    view_books()
elif choice == "3":
    search_book()
elif choice == "4":
    save_to_file()
    print("Exiting program. Thank you!")
    break
else:
    print("Invalid choice. Please select between 1 and 4.")
```

==== Library Management Menu ====

1. Add Book
2. View Books
3. Search Book by Category
4. Save & Exit

--- Library Book Records ---

ID: 1

Title: Python Programming

Author: John Doe

Category: PROGRAMMING

ID: 2

Title: Data Structures

Author: Mark Allen

Category: COMPUTER SCIENCE

ID: 3

Title: Machine Learning

Author: Andrew Ng

Category: AI

ID: 4

Title: Deep Learning

Author: Ian Goodfellow

Category: AI

ID: 5

Title: Operating Systems

Author: Silberschatz

Category: COMPUTER SCIENCE

ID: 6

Title: Database Management

Author: Elmasri

Category: DATABASE

ID: 7

Title: Computer Networks

Author: Tanenbaum

Category: NETWORKING

ID: 8

Title: Web Development

Author: Tim Berners-Lee

Category: WEB

ID: 9

Title: Cyber Security

Author: Bruce Schneier

Category: SECURITY

ID: 10

Title: Data Science

Author: Joel Grus

Category: DATA

==== Library Management Menu ====

1. Add Book

2. View Books

3. Search Book by Category

4. Save & Exit

Library records saved successfully!

Exiting program. Thank you!

In []: